

Local sieves, correlations, and witness counts for shifted even squares

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Abstract

For the shifted even square $G_a(m) = 4m^2 + a$, we count residue classes avoiding a finite set of prime divisors or prime-square divisors. Quadratic residues and the Chinese remainder theorem give exact densities, including joint densities for several shifts at the same center. We classify the roots modulo p^2 by an elementary lifting argument and obtain explicit local counts for the opposite-sign pair $4m^2 \pm q$, with q an odd prime. This pair has no local obstruction for either finite sieve. Exact computations implement the formulas; a separate factorization experiment compares prime witnesses with rough squarefree almost-prime witnesses. Finite moment inequalities relate the number of witnesses to the number of occupied centers. The results concern finite local sieves and do not establish global squarefreeness or simultaneous prime values.

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1 Problem and notation

For an even positive integer $n = 2m$ and an odd prime q , consider

$$n^2 + q = 4m^2 + q, \quad n^2 - q = 4m^2 - q.$$

Before asking whether these values are prime, one can ask which centers are ruled out by a particular divisor. That question is periodic and can be answered exactly. The same idea works for prime squares and provides a finite local test towards squarefreeness.

Definition 1.1. For an integer a , put $G_a(m) = 4m^2 + a$. For $d \geq 1$, define

$$R_a(d) = \{r \in \{0, \dots, d-1\} : d \mid G_a(r)\}, \quad \rho_a(d) = |R_a(d)|.$$

Let $\mathcal{P}$ be a finite set of distinct primes, and write

$$P = \prod_{p \in \mathcal{P}} p, \quad M = \prod_{p \in \mathcal{P}} p^2 = P^2.$$

For a finite set of integer offsets $\mathcal{A}$, a prime-sieve survivor is a center for which $p \nmid G_a(m)$ for all $a \in \mathcal{A}$ and $p \in \mathcal{P}$. A prime-square-sieve survivor instead requires $p^2 \nmid G_a(m)$ for every such a, p .

All products in this note are finite. An empty product is 1; an empty offset family imposes no restrictions. Prime-sieve survival has period P , and prime-square-sieve survival has period M . These are valid periods, not necessarily the smallest ones. The local statements allow negative centers and arbitrary integer offsets, including zero.

There are two counting questions in the project. Fixing offsets and varying m gives a local sieve count. Fixing n and searching over prime offsets q gives a witness count. The experiments measure both, but the two counts have different quantifiers. In particular, saying that both signs have a witness allows different offsets on the two sides; asking for the same q to work twice is stronger.

Definition 1.2. For $v > 1$, write $v = \prod_p p^{e_p}$ and $\Omega(v) = \sum_p e_p$. A prime witness is a prime value $v = n^2 \pm q$. A rough squarefree P_2 witness is a value $v > 1$ with every $e_p \leq 1$, $\Omega(v) \leq 2$, and least prime divisor $P^-(v) > n^{1/4}$.

Here P_2 means at most two prime factors, counted with multiplicity. Primes can qualify, whereas prime squares cannot. The program tests roughness by $(P^-(v))^4 > n$, with exact integers. Values at most 1 are in neither witness class. These are conditions on the fully factored value, stronger than passing a fixed local sieve.

For an odd prime p , the Legendre symbol $\left(\frac{t}{p}\right)$ is 0 if $p \mid t$, 1 if t is a nonzero square modulo p , and -1 otherwise. This definition is enough for the root counts below.

2 Local divisibility modulo primes

Proposition 2.1 (Prime root count). *For an integer a and an odd prime p ,*

$$p \mid G_a(m) \iff (2m)^2 \equiv -a \pmod{p}, \quad \rho_a(p) = 1 + \left(\frac{-a}{p}\right).$$

At 2, the count is 0 for odd a and 2 for even a .

Proof. Multiplication by 2 is a bijection modulo an odd prime. The equation $x^2 = 0$ has just the root 0 in the field $\mathbb{F}_p$. A nonzero square has two distinct roots $x, -x$: any other root y would satisfy $(y-x)(y+x) = 0$, so $y = x$ or $y = -x$. A nonsquare has no root by definition. Finally, $4m^2 + a \equiv a \pmod{2}$ for both classes of m . $\square$

The usual description “zero or two roots” therefore requires $p \nmid a$. For $a = \pm q$ and $p = q$, the single bad class is $m \equiv 0 \pmod{q}$. At 2, neither odd offset removes a center.

Proposition 2.2 (Squarefree moduli). *For squarefree $d \geq 1$,*

$$\rho_a(d) = \prod_{p|d} \rho_a(p).$$

If a is odd, this is at most $2^{\omega(d)}$, where $\omega(d)$ counts the distinct prime factors. At $d = 1$, the count is 1.

Proof. The Chinese remainder theorem (CRT) bijects classes modulo d with tuples of classes modulo its prime factors. Under this bijection, the roots form the Cartesian product of the local root sets. Taking cardinalities proves the formula and the preceding proposition gives the bound. Modulo 1, the sole canonical class is 0. $\square$

CRT combines coprime moduli. It does not treat repeated prime factors as independent, or different offsets at one prime as independent.

3 Finite prime sieves and interval counts

For integer endpoints $L \leq U$, put $H = U - L$ and

$$A_a(d; L, U) = \#\{m : L < m \leq U, d \mid G_a(m)\}.$$

Lemma 3.1 (Count in one residue class). *For $d \geq 1$ and $0 \leq r < d$, the number of integers in $(L, U]$ congruent to r modulo d is*

$$\left\lfloor \frac{U - r}{d} \right\rfloor - \left\lfloor \frac{L - r}{d} \right\rfloor.$$

It differs from H/d by at most 1. Hence

$$\left| A_a(d; L, U) - H \frac{\rho_a(d)}{d} \right| \leq \rho_a(d).$$

Proof. Write an integer in the class as $r + kd$ and count k between the two endpoints. Each floor differs from its argument by a number in $(-1, 0]$, giving the error bound. Sum over the roots in $R_a(d)$. $\square$

Write $S_a(L, U; P)$ for the prime-sieve survivor count of one shift, and put $\delta_a(P) = \prod_{p|P} (1 - \rho_a(p)/p)$.

Theorem 3.2 (Finite inclusion-exclusion). *For P a product of distinct primes,*

$$S_a(L, U; P) = \sum_{d|P} \mu(d) A_a(d; L, U), \quad \mu(d) = (-1)^{\omega(d)}.$$

Moreover,

$$|S_a(L, U; P) - H\delta_a(P)| \leq \prod_{p|P} (1 + \rho_a(p)).$$

For odd a and k odd sieving primes, the bound is at most 3^k .

Proof. For each m , expand the finite product

$$\mathbf{1}_{\gcd(G_a(m), P)=1} = \prod_{p|P} (1 - \mathbf{1}_{p|G_a(m)}) = \sum_{d|P} \mu(d) \mathbf{1}_{d|G_a(m)}.$$

Summing gives the identity. Lemma 3.1 bounds the total error by $\sum_{d|P} \rho_a(d)$. The CRT factorization gives

$$\sum_{d|P} \rho_a(d) = \prod_{p|P} (1 + \rho_a(p)), \quad \sum_{d|P} \frac{\mu(d) \rho_a(d)}{d} = \prod_{p|P} \left(1 - \frac{\rho_a(p)}{p}\right).$$

For odd a , the factor at 2 is 1, and each other error factor is at most 3. $\square$

Adding primes excludes more divisors, but this elementary error bound grows. It can exceed the interval length. For short windows, the exact residue-class count will be more useful than replacing it by a density.

4 Several offsets at the same center

Lemma 4.1 (Overlap of bad classes). *For integers a, b and any positive modulus d , the sets $R_a(d)$ and $R_b(d)$ are identical if $a \equiv b \pmod{d}$, and disjoint otherwise.*

Proof. Congruent offsets define the same congruence. A common root would give $d \mid G_a(m) - G_b(m) = a - b$. $\square$

Proposition 4.2 (Local covariance). *Choose m uniformly modulo a prime p and let $I_a = \mathbf{1}_{p|G_a(m)}$. Then*

$$\text{Cov}(I_a, I_b) = \begin{cases} \frac{\rho_a(p)}{p} \left(1 - \frac{\rho_a(p)}{p}\right), & a \equiv b \pmod{p}, \\ -\frac{\rho_a(p)\rho_b(p)}{p^2}, & a \not\equiv b \pmod{p}. \end{cases}$$

Proof. The expectations of I_a, I_b are $\rho_a(p)/p, \rho_b(p)/p$. By Lemma 4.1, the expectation of $I_a I_b$ is $\rho_a(p)/p$ for congruent offsets and zero otherwise. Subtract the product of the expectations; in the first case the two root counts agree. $\square$

For $a = 5, b = 11, p = 3$, both root sets are $\{1, 2\}$ and the covariance is $2/9$. Distinct offset residues with nonempty root sets instead have negative covariance. These probabilities are on a finite residue space; they are not probabilities of prime events on the integers.

Theorem 4.3 (Joint prime-sieve density). *For a finite offset family $\mathcal{A}$, let*

$$\beta_{\mathcal{A}}(p) = \left| \bigcup_{a \in \mathcal{A}} R_a(p) \right| = \sum_{c \in \mathcal{A} \bmod p} \rho_c(p),$$

where the sum uses one representative of each distinct offset residue. The number of classes modulo P surviving for every offset is

$$N_{\mathcal{A}}(P) = \prod_{p|P} (p - \beta_{\mathcal{A}}(p)), \quad \delta_{\mathcal{A}}(P) = \frac{N_{\mathcal{A}}(P)}{P} = \prod_{p|P} \left(1 - \frac{\beta_{\mathcal{A}}(p)}{p}\right).$$

Proof. Lemma 4.1 makes the root sets of distinct offset residues disjoint, proving the sum formula. There are $p - \beta_{\mathcal{A}}(p)$ good local classes. CRT identifies the global good classes with their Cartesian product. Divide the resulting count by P . $\square$

For two offsets, the bad count is $\rho_a(p)$ when $p \mid a - b$, and $\rho_a(p) + \rho_b(p)$ otherwise. Thus for $\{5, 11\}$ at 3, the joint good proportion is $1/3$, whereas multiplying the separate good proportions would incorrectly give $1/9$. Across distinct primes the product is exact because CRT supplies the bijection. Within one prime, the root union must be counted first.

5 Local squarefreeness modulo prime squares

The prime sieve removes every multiple of a selected prime. To test squarefreeness locally, we should instead remove only the multiples of its square. This changes the local problem: a root modulo p need not lift, and a repeated root can have many lifts.

Proposition 5.1 (Prime-square root count). *For an odd prime p and an integer a ,*

$$\rho_a(p^2) = \begin{cases} 1 + \left(\frac{-a}{p}\right), & p \nmid a, \\ 0, & p \mid a, \quad p^2 \nmid a, \\ p, & p^2 \mid a. \end{cases}$$

At $p = 2$, the count $\rho_a(4)$ is 4 if $4 \mid a$ and 0 otherwise.

Proof. At an odd prime, $x = 2m$ is bijective modulo p^2 , so the equation is $x^2 \equiv -a \pmod{p^2}$.

Suppose $p \nmid a$. Each root x_0 modulo p is nonzero. Choose its representative $0 \leq x_0 < p$ and write $x_0^2 + a = pc$. Every lift has the form $x = x_0 + pt$ with $0 \leq t < p$. Expanding and dividing by p shows that it is a root modulo p^2 exactly when

$$c + 2x_0t \equiv 0 \pmod{p}.$$

The coefficient $2x_0$ is invertible, so there is exactly one t . Every root modulo p^2 reduces to a root modulo p , and different roots have different reductions. The count is therefore unchanged.

If $p \mid a$, reduction modulo p forces $p \mid x$. Thus $p^2 \mid x^2$, and the original congruence can hold only if $p^2 \mid a$. This proves the zero count when p divides a just once. When $p^2 \mid a$, the roots are precisely $x = pk$ modulo p^2 , with $0 \leq k < p$, giving p roots.

Finally, $4m^2 + a \equiv a \pmod{4}$ for all four classes of m . $\square$

For example, G_3 has one root modulo 3 but none modulo 9; G_9 has the three roots 0, 3, 6 modulo 9. The zero offset falls in the last odd-prime case, and at 2 every class is bad. An even offset such as $a = 2$ has every class bad modulo 2 but no bad class modulo 4.

Theorem 5.2 (Joint finite prime-square sieve). *For a finite family of integer offsets, define*

$$\beta_{\mathcal{A}}^{(2)}(p) = \left| \bigcup_{a \in \mathcal{A}} R_a(p^2) \right| = \sum_{c \in \mathcal{A} \bmod p^2} \rho_c(p^2).$$

The number of classes modulo M for which every G_a avoids every selected prime square is

$$N_{\mathcal{A}}^{(2)}(M) = \prod_{p \in \mathcal{P}} (p^2 - \beta_{\mathcal{A}}^{(2)}(p)), \quad \delta_{\mathcal{A}}^{(2)}(M) = \frac{N_{\mathcal{A}}^{(2)}(M)}{M} = \prod_{p \in \mathcal{P}} \left(1 - \frac{\beta_{\mathcal{A}}^{(2)}(p)}{p^2} \right).$$

In particular, the density is positive exactly when $\beta_{\mathcal{A}}^{(2)}(p) < p^2$ for every selected prime.

Proof. Apply Lemma 4.1 with $d = p^2$. Offsets congruent modulo p^2 give identical bad sets, while distinct residues give disjoint bad sets. This proves the local sum, including the case $p = 2$. The moduli p^2 for different primes are pairwise coprime, so CRT multiplies the numbers of good classes. Each factor is nonnegative, and a finite product is positive exactly when each factor is positive. $\square$

For one offset the density is $\prod_{p \in \mathcal{P}} (1 - \rho_a(p^2)/p^2)$. Both empty-set conventions agree with the theorem: with no primes the period is 1 and its one class survives; with no offsets every class survives. Repeating an offset adds no restriction. Offsets congruent modulo p can nevertheless be distinct modulo p^2 : 5 and 11 coincide modulo 3, but their bad sets modulo 9 are disjoint, each with two roots.

The test here is $p^2 \nmid G_a(m)$, not $\gcd(G_a(m), M) = 1$. The latter would forbid p itself and would simply repeat the prime sieve. For instance, $G_3(3) = 39$ survives the square-of-3 test despite being divisible by 3.

6 The opposite-sign pair

The family $\{q, -q\}$ permits an explicit comparison of the two sieves. The prime q behaves differently from every other prime.

Theorem 6.1 (No local obstruction for either sieve). *Let q be an odd prime and $\mathcal{A} = \{q, -q\}$. Its local bad counts are*

p	$\beta_{\mathcal{A}}(p)$	$\beta_{\mathcal{A}}^{(2)}(p)$
$p = 2$	0	0
$p = q$	1	0
$p \neq q, p \equiv 3 \pmod{4}$	2	2
$p \neq q, p \equiv 1 \pmod{4}$	$2 + 2 \left(\frac{q}{p}\right)$	$2 + 2 \left(\frac{q}{p}\right)$

Thus $\beta_{\mathcal{A}}(p) < p$ and $\beta_{\mathcal{A}}^{(2)}(p) < p^2$ for every prime. Both finite survivor densities are positive for every finite prime set.

Proof. At 2, both offsets are odd, so there are no roots modulo 2 or 4. At q , both offsets give the root 0 modulo q ; neither offset is divisible by q^2 , so neither polynomial has a root modulo q^2 .

At an odd prime $p \neq q$, the offsets are distinct modulo p and modulo p^2 , since $p \nmid 2q$. Their root sets are disjoint at both moduli. Also $p \nmid q$, so the individual counts are unchanged under lifting. At either modulus the union count is

$$2 + \left(\frac{q}{p}\right) + \left(\frac{-q}{p}\right).$$

The identities $\left(\frac{-q}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{q}{p}\right)$ and $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ give the table.

These identities follow from Euler's criterion, which can be seen directly here. The squaring map on $\mathbb{F}_p^\times$ has kernel $\{1, -1\}$ and hence $(p-1)/2$ distinct square values. By Fermat's identity these are all roots of $x^{(p-1)/2} - 1$; every other nonzero element is a root of $x^{(p-1)/2} + 1$. Thus $t^{(p-1)/2}$ represents the Legendre symbol, and exponentiation respects multiplication. Substituting $t = -1$ gives its sign.

For $p \equiv 3 \pmod{4}$, $2 < p$, including $p = 3$. For $p \equiv 1 \pmod{4}$, $p \geq 5$ and the count is at most $4 < p$. The cases 2, q are also strict. The square counts are at most 4 at the remaining odd primes, less than p^2 . Theorems 4.3 and 5.2 now give positivity. $\square$

In particular, the prime-square density can be written

$$\delta_{\{q,-q\}}^{(2)}(M) = \prod_{\substack{p \in \mathcal{P} \\ p \neq 2, q}} \left(1 - \frac{2 + \left(\frac{q}{p}\right) + \left(\frac{-q}{p}\right)}{p^2} \right).$$

No factor is needed at 2 or q . At q , prime-square survival allows all centers, whereas prime survival removes $m \equiv 0 \pmod{q}$.

Corollary 6.2 (A wider prime-square admissible family). *Every finite family of nonzero squarefree integer offsets has positive density under every finite prime-square sieve. In particular, this holds for finite families of signed prime offsets.*

Proof. No prime square divides any offset a in such a family. Consequently, $m = 0$ is good modulo p^2 for every offset and every prime p . Each local good set is nonempty, so Theorem 5.2 applies. $\square$

Indeed, every multiple of P is such a prime-square survivor, since $G_a(m) \equiv a \pmod{p^2}$ for each selected p . This does not make larger signed-prime families admissible for the prime sieve: $\{3, 5, 7\}$ already obstructs it at 3.

Each positive finite density supplies a surviving residue class, which repeats indefinitely. This guarantees centers surviving any one fixed finite sieve. It says nothing about surviving all prime squares simultaneously or producing simultaneous primes.

7 Exact implementation and interval evaluation

The code constructs local roots, takes the union over offsets, and combines the good classes by CRT. For prime squares, a nondegenerate root r of G_a modulo p is lifted directly in the m variable:

$$c = \frac{4r^2 + a}{p}, \quad t \equiv -c(8r)^{-1} \pmod{p}, \quad m = r + pt.$$

Here $8r$ is invertible when p is odd and $p \nmid a$. This follows by expanding $G_a(r + pt)$; it is the same one-step argument as Proposition 5.1. The singular cases and modulus 4 are constructed separately.

Suppose r is good modulo the current period D and s is good modulo a coprime next modulus h , either p or p^2 . The combined canonical class is

$$r + D((s - r)D^{-1} \bmod h).$$

The parenthesized residue lies in $\{0, \dots, h - 1\}$, so the result is in $[0, Dh)$ and has the required two residues. Each pair gives one distinct class, implementing the CRT bijection.

Proposition 7.1 (Exact interval evaluation). *Let $D \geq 1$ be a period and $\mathcal{R} \subseteq \{0, \dots, D - 1\}$ its survivor classes, with $N = |\mathcal{R}|$. Then*

$$S(L, U) = \sum_{r \in \mathcal{R}} \left(\left\lfloor \frac{U - r}{D} \right\rfloor - \left\lfloor \frac{L - r}{D} \right\rfloor \right).$$

If $H = tD + h$, $0 \leq h < D$, then

$$S(L, U) = tN + T, \quad 0 \leq T \leq \min(h, N).$$

In particular $|S(L, U) - HN/D| \leq N$, with equality of the count and main term whenever $D \mid H$.

Proof. Sum Lemma 3.1 over $\mathcal{R}$. Partition the interval into t consecutive blocks of D integers and a remainder of h integers. Every full block contains each residue once; the remainder contains no repeated residue and at most $\min(h, N)$ good ones. Both T and hN/D lie in $[0, N]$, giving the error bound. If $h = 0$, both are zero. $\square$

Take $D = P$ for the prime sieve and $D = M$ for the prime-square sieve. For one shift, either this bound or the inclusion-exclusion bound can be smaller. The program uses the exact floor sum rather than rounding a density times the interval length.

The reusable routines are in `src/arithmetic.py`, `src/sieve.py`, and `src/squarefree.py`. Moduli are checked to be distinct primes. Empty prime sets use the period 1. Local counts and densities use integers and rational arithmetic. Direct divisibility enumeration checks the constructed classes and interval counts independently. These checks verify the implementation; the proofs establish the formulas. Explicit class storage grows with the period and is intended for modest products. The repository's historical name `quadratic-sieve-lab` does not refer to an implementation of the classical quadratic-sieve factorization algorithm.

8 Divisor compatibility and witness moments

The local sieve fixes offsets. For the separate witness problem, two elementary tools help describe collisions and occupancy.

Proposition 8.1 (Divisor compatibility). *Let d, e be positive integers with $d \mid G_a(m)$. Write $g = \gcd(d, e)$, $d = gu$, $e = gv$, and $k = G_a(m)/d$. Then*

$$e \mid G_b(m) \iff g \mid a - b \text{ and } uk \equiv (a - b)/g \pmod{v}.$$

In particular, simultaneous divisibility requires $\gcd(d, e) \mid a - b$.

Proof. Subtraction gives $G_b(m) = guk - (a - b)$. If e divides this value, then $g \mid a - b$. Under that condition, dividing by g leaves exactly $v \mid uk - (a - b)/g$. Reverse the steps for sufficiency. $\square$

The gcd condition alone is insufficient: with $a = 3, b = 5, m = 1, d = 7, e = 5$, we have $d \mid G_a(1)$ and $\gcd(d, e) = 1 \mid a - b$, but $5 \nmid G_b(1) = 9$. Equal signs involve $q - r$ up to sign; opposite signs involve $q + r$ up to sign. No squarefreeness assumption on d, e is needed.

Let $\mathcal{C}$ be a finite set of H centers. For one sign and witness class, let $\nu(n)$ count successful offsets and define

$$M_1 = \sum_{n \in \mathcal{C}} \nu(n), \quad M_2 = \sum_{n \in \mathcal{C}} \nu(n)^2, \quad F_2 = \sum_{n \in \mathcal{C}} \nu(n)(\nu(n) - 1), \quad S = \#\{n \in \mathcal{C} : \nu(n) > 0\}.$$

The factorial moment F_2 counts ordered pairs of distinct successful offsets at the same center.

Proposition 8.2 (Moments and occupancy). *Always $M_2 = M_1 + F_2$ and $S \leq \min(H, M_1)$. If $M_2 > 0$, then*

$$\frac{M_1^2}{M_2} \leq S, \quad 0 \leq M_1 - S \leq \frac{F_2}{2}.$$

If $M_2 = 0$, all counts vanish; the lower bound is then interpreted as 0.

Proof. For a nonnegative integer j , $j^2 = j + j(j - 1)$. Every occupied center contributes at least one to M_1 . Cauchy–Schwarz on the occupied centers gives $M_1^2 \leq SM_2$. Finally, $0 \leq j - \mathbf{1}_{j>0} \leq j(j - 1)/2$: both sides vanish at $j = 0$, and for $j \geq 1$ this is $j - 1 \leq j(j - 1)/2$. Sum these inequalities. If $M_2 = 0$, every $\nu(n)^2$ is zero. $\square$

If one separately proved $M_1 \geq cHK$ and $M_2 \leq CH(K+K^2)$ with $c, C, K > 0$, this would imply

$$S \geq \frac{c^2}{C} H \frac{K}{1+K}.$$

The moment hypotheses at unbounded scales are not proved here.

For the sets $A_+, A_- \subseteq \mathcal{C}$ occupied on each sign, inclusion-exclusion gives

$$|A_+ \cap A_-| \geq \max(0, |A_+| + |A_-| - H).$$

Two disjoint sets can each occupy 40% of the centers. Positive separate occupancy need not force overlap. Even when $A_+ \cap A_-$ is nonempty, the two witnesses can use different offsets.

9 Exact finite experiments

9.1 Fixed-offset prime and prime-square sieves

For $\mathcal{P} = \{2, 3, 5, 7\}$, the prime period is $P = 210$ and the prime-square period is $M = 44,100$. Both experiments construct CRT classes, check a complete period by direct enumeration, and count the interval $500 < m \leq 1000$. The following tables are exact finite counts.

Offsets	Classes mod 210	Density	Interval count
$\{3\}$	100	10/21	237
$\{-3\}$	140	2/3	333
$\{3, -3\}$	100	10/21	237
$\{5\}$	40	4/21	95
$\{-5\}$	168	4/5	400
$\{5, -5\}$	40	4/21	95
$\{5, 11\}$	20	2/21	47

Offsets	Classes mod 44,100	Density	Interval count
$\{3\}$	42,300	47/49	480
$\{-3\}$	44,100	1	500
$\{3, -3\}$	42,300	47/49	480
$\{5\}$	32,900	47/63	372
$\{-5\}$	44,100	1	500
$\{5, -5\}$	32,900	47/63	372
$\{5, 11\}$	21,620	1081/2205	246
$\{9\}$	27,048	46/75	305

For $\{5, -5\}$, the prime-square bad counts at 2, 3, 5, 7 are 0, 2, 0, 2, giving $(1 - 2/9)(1 - 2/49) = 47/63$. For $\{5, 11\}$, the bad count at 3^2 is 4, despite the coincident root sets modulo 3. These examples show both the change from prime to prime-square conditions and the need to group offsets at the correct modulus. Density 1 for $\{-5\}$ refers only to these four selected squares.

The prime-square experiment also includes offsets 0, 2, 4, repeated offsets and an empty family. Tests cover empty prime sets, singular roots, negative intervals and invalid moduli. The interval length is 500, not a multiple of either period, so an interval count need not equal 500 times its period density.

9.2 Witnesses with varying prime offsets

The original experiment tests the 500 even centers $1000 < n \leq 2000$ and the 13 odd primes $3 \leq q \leq 43$ on both signs, giving $500 \cdot 13 \cdot 2 = 13,000$ values. Trial division fully factors each positive value. Every factor is checked for primality and the product is checked against the input. Failures are retained alongside successes.

R	Witness class	Upper occupied	Lower occupied	Both sides
7	Prime	216	236	104
7	Rough squarefree P_2	454	461	421
19	Prime	329	380	249
19	Rough squarefree P_2	497	498	495
43	Prime	439	448	390
43	Rough squarefree P_2	499	500	499

The last column allows different offsets on the two signs. At $R = 7$, upper prime witnesses have $M_1 = 246$, $M_2 = 306$, $F_2 = 60$, and $S = 216$. Thus the lower bound is $246^2/306 = 3362/17$, and $M_1 - S = 30 = F_2/2$. The saved summary includes moments for all twelve sign/class/cutoff combinations. Rough squarefree P_2 occupancy is much higher in this window; it cannot be relabeled as prime occupancy.

9.3 Reproduction

From the repository root, with Python 3.10 or later and no third-party packages, run

```
python -m unittest discover -s tests -v
python -m experiments.witness_counts
python -m experiments.local_sieve
python -m experiments.squarefree_sieve
```

The witness experiment writes `results/witnesses.csv` and `results/summary.json`. The two sieve experiments each write a JSON record and a Markdown table in `results/`. The checked-in default results correspond to the tables above. Each module accepts `-help` and `-output`; the latter keeps alternative runs separate. The legacy entry point `experiment.py` still runs the witness experiment.

Tests compare every offset residue modulo p^2 for primes through 19 with brute-force roots, verify overlaps, and compare small CRT periods and intervals with direct divisibility tests. The original 272 root/count checks and 47,670 quotient checks are retained. These ranges describe software verification, not the scope of the proofs.

10 Limitations and status

A finite sieve survivor need not be prime or globally squarefree. For example, $G_5(2) = 21$ avoids the primes 2, 5, 11 but is composite. Also $G_9(2) = 25$ avoids the prime squares 4, 9 but is not squarefree. Conversely, a prime value equal to a sieving prime is removed: $G_{-11}(2) = 5$ fails the prime sieve containing 5.

For a positive value $v > 1$, checking every prime up to $\sqrt{v}$ does certify primality if no divisor is found: a composite value has a prime divisor at most $\sqrt{v}$. Checking prime-square divisibility for

every prime up to $\sqrt{v}$ likewise certifies squarefreeness. Both are bounded tests for a specified value, rather than consequences of one fixed finite prime set as m grows.

The CRT densities are exact for periodic local survivor sets. They are not densities of prime values or of values avoiding every prime square. Positivity for every finite prime set does not by itself justify passing to an infinite set of restrictions. The opposite-sign theorem proves local admissibility and supplies no global simultaneous-squarefreeness or simultaneous-prime theorem.

The witness experiment tests exact factorization, squarefreeness, the P_2 condition and roughness. A finite prime-square sieve alone does not supply those properties. Its tables concern the stated window and cutoffs, and the moment inequalities require separately proved moment estimates for any asymptotic deduction. No estimate uniform in a growing sieve period is claimed.

The project uses elementary tools: quadratic residues, CRT, finite inclusion-exclusion, and Cauchy–Schwarz. The development here is their combination for several shifted squares, the extension to prime-square local conditions, and an exact implementation. Proofs of the stated results are included.

11 Further questions

1. Higher prime powers. Determine $\rho_a(p^k)$ for $k \geq 3$, separating the possible orders of divisibility of a . The nondegenerate one-step lift can be iterated; the repeated-root cases require a separate count. Compare finite power-free sieves and the exact periods of their survivor sets.

2. Larger families and prime obstructions. Classify minimal signed-prime-offset families with $\beta_{\mathcal{A}}(p) = p$, beginning at 3 and 5. Such families obstruct the prime sieve even though Corollary 6.2 excludes prime-square obstructions. The criterion is whether the offset residues cover all values $-4m^2$ modulo p ; the challenge is to give a useful classification.

3. Discrepancy under translation. For one fixed survivor set of period D and a fixed length H , determine the maximum and minimum of $S(L, L + H) - HN/D$ as L runs modulo D . The remainder count in Proposition 7.1 gives a starting bound. Can the arrangement of good classes improve it for these sieves? This is a finite combinatorial question for either kind of sieve.